

Table 1. Average contact angle measurements for each surface studied.

	static contact angle	receding contact angle	advancing contact angle	contact angle hysteresis
smooth copper	52°	~10°	66°	56°
smooth PDMS	106°	~10°	110°	100°
smooth Teflon	123°	111°	129°	18°
rose petal	145°	~30°	154°	~120°
parahydrophobic PDMS	144°	~30°	161°	~130°
superhydrophobic PDMS	151°	147°	156°	9°
parahydrophobic Teflon	135°	105°	146°	41°

2.3. Experimental Design

Boiling experiments were carried out in a closed-chamber pool boiling facility (Figure 2(a)). The facility and experimental procedure, previously described in Ref. [34], are briefly summarized here. The facility consists of a closed boiling chamber, an external Graham condenser used during degassing, two internal immersion heaters to maintain saturated conditions, an internal coiled condenser to maintain constant pressure during sealed operation, and temperature and pressure sensors to measure the conditions inside the chamber. Prior to the boiling test, the chamber is filled with deionized water which is then